

Unit 6

Atmosphere



Chapter Outline

6.1 Introduction
6.2 Composition of the Atmosphere
6.3 Temperature and Heat Budget
6.4 Atmospheric Pressure and Winds
6.5 Humidity, Condensation and Clouds
6.6 Precipitation
6.7 Atmospheric Disturbances (Cyclone and Anti Cyclone)

Learning Objectives:

Students must be able to

- *Understand the composition and nature of atmospheric layers.*
- *Understand the vertical and horizontal distribution of temperature in the atmosphere.*
- *Explain the mechanism of formation of various wind systems of the world.*
- *Identify various forms and types of precipitation.*

6.1 Introduction

You must have heard people, in the countryside, saying

"When sheep collect and huddle,
Tomorrow will puddle!"
"If ants march in a straight line,
expect rain"

Phrases like 'a cold morning', 'sunny day', 'cloudy day' and rainy day refer to the weather. Weather refers to the state of atmosphere at a particular place at any given time denoting the short term variations of atmosphere in terms of temperature, pressure, wind, moisture, cloudiness, precipitation and other elements. Weather is highly variable from time to time, day to day and place to place. Weather is not constant. It is always changing within hours or a day.

On the other hand, climate is the average weather conditions of an area for a long period of time. The World Meteorological Organisation (WMO) has suggested data for a period of 30 consecutive years to be referred for calculating the climatic averages of various weather elements. Climate is constant. It is a permanent condition of a place.

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